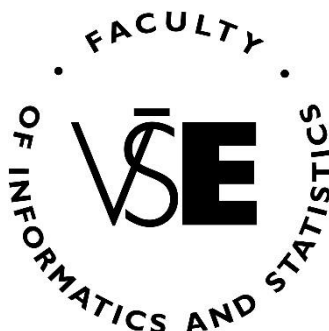


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**VIDEOGAME SKINS AS AN ALTERNATIVE
INVESTMENT INSTRUMENT: THE CASE OF
COUNTER STRIKE: GLOBAL OFFENSIVE**

BACHELOR THESIS

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I hereby declare on my honor that I wrote this bachelor's thesis independently, and I used no other sources and aids than those indicated.

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Prague, 28.4.2023

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Abstract

This thesis deals with the game Counter Strike: Global Offensive in which there are collectable in-game items. The research consists of testing whether there is a relationship in the development of price of CS:GO items, specifically the loot box Operation Breakout Case and the Redline Field-Tested skin on AK-47 weapon, and the development of price of more traditional investment instruments. The examined period is 2020, in which the investment instruments were more volatile. Using the least square method, two regression analysis models are created from the data to further examine the price dependencies of the selected investment instruments. In the first model, in case of the Operation Breakout Case, there is no indication of any relationship in the price development with other investment instruments. In the second model, in the examination of the AK-47 Redline a dependency in price development with Bitcoin is found. However, this does not refute the results of the first model in any way, but rather confirms the necessity of researching the potential of individual items and not as a whole. Thus, the thesis shows the potential of CS:GO items entering an investors riskier portfolio as an alternative investment instrument, as these collectables might have a unique development in price, but on the other hand, they are still not researched in detail.

Keywords: CSGO, skins, investment instruments, market price, alternative investing

Abstrakt

Tato práce se zabývá hrou Counter Strike: Global Offensive ve které jsou sběratelské herní předměty. Zkoumání spočívá formou testování, zdalipak existuje vztah ve vývoji cen předmětů z CS:GO, konkrétně bedna Operation Breakout Case a skin na zbraň AK-47 Redline Opotřebený a vývojem cen tradičnějších investičních instrumentů. Zkoumaným obdobím je rok 2020, ve kterém byly investiční instrumenty volatilnější. Použitím metody nejmenších čtverců pak jsou z dat vytvořeny dva modely regresní analýzy, kde se dále zkoumají závislosti cenového vývoje vybraných investičních instrumentů. V prvním modelu, v případě Operation Breakout Case, nejsou ani známky náznaku existence vztahu ve vývoji cen s ostatními investičními instrumenty. Ve druhém modelu, při zkoumání AK-47 Redline, je nalezena závislost ve vývoji ceny s Bitcoinem. To ovšem nijak nevyvrací výsledky prvního modelu, ale naopak potvrzuje nutnost zkoumání potenciálu jednotlivých předmětů, a nikoliv jako celku. Práce ukazuje potenciál vstupu CS:GO předmětů jakožto alternativního investičního instrumentu do rizikovějšího portfolia investora, jelikož tyto sběratelské předměty můžou mít unikátní vývoj, ovšem na druhou stranu, stále nejsou ještě podrobně prozkoumané.

Klíčová slova: CS:GO, skiny, investiční instrumenty, tržní cena, alternativní investování

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Introduction

Over the last few years, video games market size grew significantly and it is predicted to grow even more in the future (Kirkcaldy, 2023). There are reasons for the growing popularity in games, one of which is competitiveness and e-sports. With the rising demand in games, there is also an expectation of rising demand for in-game items or collectibles in already existing popular games. Most games do not have a trading system or marketplace and valuing these items is difficult. However, there are games that do have marketplace and trading system. This allows valuing the items based on their supply and demand. This schema is also used by some traditional investment instruments such as stocks, gold, or crypto currencies. One of the biggest games which falls into this category is Counter Strike: Global Offensive (or CS:GO).

The goal of this thesis is to make a step forward in the investing in CS:GO in-game items, by understanding the development of selected items in comparison to traditional commodities. The examined period is from January 2020 to end of December 2020. The main reason behind the selection of this period were covid restrictions. During this time stock market and prices of other commodities were highly volatile (Frazier, 2021). Also during that time player base of CS:GO grew (SteamCharts, 2023). The real question is whether the price change of traditional investment instruments is similar or same as the selected in game items.

Theoretical part of the thesis introduces terminology connected with the rest of the thesis, about CS:GO. The core of the theoretical part is to understand all in-game items as that should provide an assistance for the decisions in what items to invest in. Since there are many ways to obtain in-game items in CS:GO, all variants will be presented. The whole theoretical part should provide information for the empirical part.

The empirical part of this thesis contains usage of linear regression models to find relations between selected in-game items from the game CS:GO and selected investment instruments. The in-game items are chosen with the assistance of theoretical part and based on their popularity. The first item is Operation Breakout Weapon Case and the second is AK-47 Redline in Field-Tested condition. Among the selected investment instruments are chosen S&P 500, Gold, and one more volatile, Bitcoin. The reasoning behind this selection is to pick two more traditional instruments and one unconventional. For each item, a model will be made. The results from the models will be analysed with the hypothesis and then interpreted.

1 Theoretical part

Around half of the population of Czechia invests in some way (Redakce, 2022). This thesis aims to analyse another potential alternative. For the analysis in the empirical part, it is necessary to understand what this new alternative is and whether it has potential for regular investors. At first it is essential to understand some background about the organisation, the game, and the items, which might have the potential to become an interesting investment instrument.

1.1 Valve

Valve is an American corporation that develops games and provides a digital distribution of software. Valve created a distribution platform Steam and franchises like Half-Life, Counter-Strike, Dota and more (Valve Corporation, 2023).

Steam platform not only allows people to buy software, but also provides many features for the community such as Steam workshop¹ and community market. Steam community market is based on supply and demand. Players can make a sell order of any item from the supported games from their Steam inventory. There are also official tools like multi sell, that allows players to make more sell orders of the same item at once. Buyer uses Steam wallet, which can be funded with standard online payment methods. Buyer can also make a buy order and just wait until the price reaches the order or when he wants to make a purchase in bulk. To make the Steam market effective a Steam transaction fee is included. The transaction fee is set to 5 % (with a minimum fee of \$0,01) and is covered by the buyer (Steamsupport, 2023). There is also a fee for the CS:GO in-game items, set to 10 %. So, the total fee per one CS:GO in-game item is 15 %.

However, Steam market also has flaws one of which is the price restriction. The maximum amount a player can have in his Steam wallet is \$2000 or similar amount in other currencies. Listing of one item cannot exceed \$1800, or when the price of the listed items combined with the current amount in the Steam wallet exceeds \$2000 (Steamsupport, 2023). This forces players to use external marketplaces, where price is not a limitation. Another shortcoming is the absence of bargain, it is not essential feature, but most third-party marketplaces do have this feature.

Another community feature is trading, which allows players to trade their items from games between each other. It is possible to only trade tradable² items from supported games. To start a trade the other side sends you their link that allows you to create the trade

¹ Steam workshop allows to share and use player-created content in supported games.

² Some items are untradable, or trade restricted for some time.

offer. As a receiver you can accept the offer, decline, or create a counteroffer (Steam Support 2023a).

To make the platform safer Valve decided to make Steam Guard. If a player wants to use Steam marketplace, he must have the Steam Guard turned on for at least 15 days (Steam Support, 2023). Steam guard is a two-step authentication in the Steam mobile application, which replaced the traditional verification via e-mail. When a player wants to make a trade, or an item on Steam market, he must confirm each action on his mobile phone. Steam Guard is also used when a player signs in to Steam as a platform or on the Steam website. Players can also be banned from the Steam marketplace or trade banned by Valve.

1.2 Counter strike: Global offensive

As mentioned previously, CS:GO is a video game developed by Valve. It was officially released on August 12, 2012. It is a team based first person-shooter with many different modes. The classic game mode is competitive (*Counter-Strike: Global Offensive* 2023). There are ten players in total that are divided into two sides terrorists and counter-terrorists. Both teams compete in a best-of-thirty match³. Each team has five players and different goals to win rounds. Since it is a first-person shooter, weapons play a key role. Some weapons are available for both teams, but some are unique to terrorists and some to counter-terrorists. All players get every default weapon for free into their inventory. Default weapons can be renamed, or a player can apply stickers on them, but they cannot be traded or sold on Steam market. However, there is a lot of item types in the game. Let us break them down.

1.2.1 Cases

Cases work on the loot box system. Loot boxes function on the basis of Random reward mechanism (Nielsen, Grabarczyk, 2019). Each case contains unique skins in different rarity and exceedingly rare item⁴ which can be obtained only from cases. A skin is a cosmetic weapon finish. For a visual presentation of what a skin is visit Appendix A. Player can open the case using a key, which can be purchased⁵ directly in the game as an in-game item that cannot be traded or sold. As a reward for the opening, he receives a random item. Any item also has a 10 % chance of being StatTrak (Çakır, 2020). StatTrak is a feature in the game that tracks number of kills of an item with this tag. Table 1.1 shows standard format or rarities and odds of obtaining each rarity.

³ Best-of-thirty match, is match with total max of thirty rounds, where the first team to get 16 rounds win or the game ends up as a tie 15-15.

⁴ Exceedingly rare item can be either a knife or gloves, for more information see chapter 1.5.

⁵ During the writing of this thesis the price of each key is \$2,5 or 2,35 €. Price changes over time and is determined by the developer Valve.

Table 1.1: Odds of obtaining an item of given rarity (William, 2022).

Rarity	Odds of obtaining
Mil-Spec (blue rarity)	79,92 %
Restricted (purple rarity)	15,98 %
Classified (pink rarity)	3,2 %
Covert (red rarity)	0,64 %
Exceedingly rare (yellow rarity)	0,25 %

Besides buying the case on the Steam market or receiving it from a trade, player can obtain one item called “drop” every week by levelling up profile rank. That can be done simply by playing the game, because after each game every player is rewarded with experience towards the level up. As a drop, player can obtain skin, graffiti, or a case (Clement, 2021).

There are three pools of cases drops. Prime drop pool, rare drop pool and discontinued drop pool. Prime drop pool has five cases that have the highest chance of being a drop for a player. When a new case comes out it is added to the prime drop pool, replacing one of the cases. Unless the case is released with some special occasion like operation. Operation is a special form of additional content including many new features for a limited time in the game. When a case is removed from drop pool it is moved to rare pool. Players can still obtain cases from rare pool, with estimated probability of 1 % (*CS:GO Cases*, 2023). Discounted pool contains cases from operations. These cases were available only during the operation and are no longer available to be obtained directly from the game.

1.2.2 Stickers and capsules

Capsules work on the same system as cases, but they do not contain skins but stickers. They also do not include exceedingly rare item. There are numerous types of sticker capsules, from community themed stickers to team logo stickers. Unlike cases, capsules cannot be obtained as a drop for free by just playing the game but must be purchased from the in-game store using Steam wallet funds. They can also be traded, purchased, or sold on Steam market. They are a part of operations, majors⁶ or for instance part of a special anniversary updates. Therefore, they are available in the in-game store for a limited period. Even though majors are usually twice a year, they always contain new sticker capsules with some changes. Stickers in sticker capsules also might have different rarity as skins in cases. Although some capsules have limited rarities, (for instance only blue and purple rarity) odds of obtaining certain rarities are similar.

⁶ Majors are Valve sponsored esports tournaments (*CS:GO Major Championships*, 2023). They are considered as the biggest type of tournament in CS:GO and include special features like team stickers, souvenir packages and in-game tributes (Stenhouse, 2016).

Stickers are used as a skin enhancement. Player can apply up to four stickers on each weapon skin. Stickers can be also scraped from the weapon skin. There are also many types of stickers such as: regular, glitter, foil, holo, gold or newly introduced lenticular⁷. Like cases, stickers and capsules are considered as consumables. Once they are used (opened or applied) the supply is lowered.

1.2.3 Collections and Souvenir packages

As mentioned previously, players can receive a skin as a part of a weekly drop. Skins from weekly drops cannot be from regular cases but are from collections. Collections follow the same rarity format with the addition of Industrial (light blue rarity) and Consumer (grey rarity) (CSGO Database, 2023). Those two rarities are more common than Mil-Spec (blue rarity). Since most skins from collections are obtained as a drop, odds have not been determined. Collections in the drop pool change over time. Some collections were available during operations, where players could get random item in an exchange for operation stars. Since these collections contain two new rarities, it is assumed that odds are shifted, where the probability of receiving grey is like blue in the standard format. With each better rarity the odds are five times lower. Collections also do not contain exceedingly rare item.

Souvenir packages contain skins from collections, with the exception that the opened skin has four gold stickers on it. There have been changes in the past, but in the most recent packages skins have two team stickers, one tournament sticker and a map sticker. Souvenir packages can be exchanged for tokens which can be purchased in-game during majors. Player chooses specific match that has been played on a map between two teams, which determines what stickers will the skin have. Since these packages can be purchased from in-game store, they do not require additional key to open them. The selection of maps changes every year. There are always seven maps in the active pool, where every map has its own collection.

1.2.4 Agents and patches

Agents are cosmetics for the player models. There are currently only three agent collections, which were all available with three most recent operations (csgoStash, 2023). They are also divided into standard rarities. Agents are not obtained from any exceptional cases. During operation, player can use operation stars⁸ to receive a random agent from chosen rarity (for instance for 25 operation stars, player receives a random agent of red rarity, or for 5 stars a random agent of blue rarity) (Shirley, 2022). They were not received well by the community and are also banned in most esports tournaments mainly due to visibility issues (Ashton, 2019).

⁷ Each type has a different effect. Since same types are different rarities in various cases, it cannot be generally divided.

⁸ Operation stars are essentially funds for the operation items. They are available to purchase for Steam wallet funds.

Players can apply patches on their agents. It works on the same principal as stickers. Patches can be obtained from patch packs which are same as sticker capsules. They follow the standard rarity system and are usually as a part of the major, operation or special event. The main downside of agents and patches is that players cannot see their own player model.

1.2.5 Other items

Besides previously mentioned items and skins which have their own chapter 1.4, there is still many other items such as music kits, collectable pins, graffiti, name tag, StatTrak swap tool, containers and operation and viewer passes.

Music kits are obtained from music kit boxes. They are available during unique events or occasions in the in-game store. Music kit boxes only have one rarity, so all kits have same odds. They also do have the 10 % chance of being StatTrak. Unlike StatTrak on weapon skins that count number of kills, music kits count number of time player has been MVP⁹. Besides changing the in-game music for the player, music kits serve no other purpose.

Collectable pins are obtained from pins capsules. These capsules cannot be purchased from in-game store. Player must attend a real-life event like major, to receive a physical pin and as a bonus he receives an in-game pin capsule (Brolaub, 2015). Collectable pins are only used as a cosmetic accessory on the CS:GO player profile.

Each graffiti can be sprayed on the wall in-game 50 times, before it expires and lasts 7 minutes (Sherif, 2016). As already mentioned, graffiti can be obtained as a weekly drop for free. There were also three graffiti boxes for a limited time in the in-game store. Once again, they follow the same rarity system without having the exceedingly rare item. Graffiti is also considered as consumable.

Name tag and StatTrak swap tool are permanently in the in-game store for a fixed price. Name tag is used to change the nickname of a weapon skin. StatTrak swap tool can swap number of kills from the same weapon type (for instance from AWP to AWP) between different weapon skins. Since these two items have fixed price by the developer, they can be used well in trading, but serve no purpose in investing.

CS:GO inventory is limited to store up to one thousand items (Steam Support, 2023). Containers (or Storage units) allows to store additional one thousand items into them. They can be purchased from in-game store for a fixed price and cannot be traded or sold even when they are empty.

Operation passes and viewer passes allows players to gain access to more features. Operation pass can be purchased during operations and viewer pass during majors. Both passes transform into pins that can be displayed on profile. These pins can be upgraded for

⁹ MVP stands for most valued player. Every round one player of the winning team is awarded with MVP, based on the biggest impact that round.

completing special challenges. Operation pass grants an access to operation stars, which can be obtained by completing special challenges or can be purchased for Steam wallet funds. Operation stars can be exchanged for an operation case, skins, agents, graffiti, or stickers. For completing special challenges during majors, players obtain souvenir tokens that can be exchanged for souvenir packages. Souvenir tokens can also be purchased for Steam wallet funds.

1.3 Skins in Counter strike: Global Offensive

CS:GO skins were added to the game as a part of the update called “Arms Deal” back in 2013 (Counter-Strike 2013). This update is considered to be the most important in the history of this game (Evan, 2015). Since then, over one thousand skins were added to the game. But one skin was also removed from the game (Savage, 2014) due to copyright issues.

Skins are cosmetic weapon finishes (illustration is in Appendix A). They only change the appearance of weapons, knives, or gloves, but do not change the gameplay at all. There are numerous skins on each weapon. There are currently thirty-four weapons in the game divided into four categories presented in table 1.2. It is obvious that from this selection only few are used regularly by players (Patdu, 2023). Which weapons are used is defined by the meta (most effective tactics available). Since stats of weapons are changed by developer in time, popularity of weapons also changes in time. All this reflects on the demand of skins for certain weapons. Especially in case of M4A4 and M4A1-S, where player must choose before the game starts which of these weapons will be available for him to purchase during the game.

Table 1.2: List of all weapons in Counter strike: Global Offensive (Valve Developer Community, 2023)

Pistols	Rifles	SMGs¹⁰	Heavy
CZ75-Auto	AK-47	MAC-10	MAG-7
Desert Eagle	AUG	MP5-SD	Nova
Dual Berettas	AWP	MP7	Sawed-Off
Five-Seven	Famas	MP9	XM1014
Glock-18	Galil AR	PP-Bizon	M249
P2000	G3SG1	P90	Negev
P250	M4A1-S	UMP-45	
R8 Revolver	M4A4		
Tec-9	SCAR-20		
USP-S	SG 553		
	SSG 08		

¹⁰ SMGs stands for sub machine guns.

Skins work as an extra painted layer on the default weapon. Whenever skin is obtained it has assigned float value to it. Float value can range from 0 to 1 and defines the skin wear. Whenever is float value defined in does not change while using the skin. The higher the float value is, the less of the paint is applied on the default weapon. In general, lower float value means better, as the skin looks cleaner. Some skins have a limited range of float value (for instance only from 0.18 to 1.00 or from 0.00 to 0.07). Based on the float value, skins have associated skin wear dividing it in five categories. Skin wear is an official tag on the skin, used on Steam market or while trading. In the table 1.3 are shown all five skin wear categories with the associated float value.

Table 1.3: Summary of skin wear to float values (Churick, 2023).

Skin wear	Float value
Factory new (FN)	0.00 – 0.07
Minimal wear (MW)	0.07 – 0.15
Field tested (FT)	0.15 – 0.37
Well worn (WW)	0.37 – 0.44
Battle scarred (BS)	0.44 – 1.00

Besides float value, each skin has another value called pattern. Pattern can range from 1 to 999 and is also defined when the skins is obtained. Most skins look the same with different pattern values. But some skins are pattern based, so the look is completely defined by the pattern.

1.3.1 Trade up contract

Besides opening cases, souvenir packages or receiving one from drops, there is one more way that influences total supply of a skin. Trade up contract is an additional feature available in players inventories. Player can trade ten skins of the same rarity for one skin of the better rarity (for instance 10 Mil-Spec (blue rarity) for 1 Restricted (purple rarity)). The traded skin is randomly chosen from the collection or collections that are included in the ten skins in trade. If all ten skins are from the same collection and there are more skins of the higher rarity, one of the skins is chosen at random. After the trade all ten skins are completely removed from the game and one new skin is created. The float value is determined by the ten traded skins, but pattern value is newly generated. You can use both StatTrak and non-StatTrak skins, but you cannot combine them. Souvenir skins cannot be used in trade up contracts. The highest rarity a player can trade is Classified (purple rarity) to get the Covert (red rarity) skin. It is not possible to trade up 10 Covert skins to obtain an Exceedingly rare item as these items can be only obtained from case opening.

1.4 Knives and gloves

Last two item types that have not been presented are knives and gloves. They can be obtained only by opening cases as an Exceedingly rare item.

Knife in the game as a weapon cannot be dropped to other players, unlike every other weapon. When a player holds a knife instead of a regular weapon he moves faster. So, at the beginning of the round and then when player changes position, he uses a knife. From authors perspective a knife is the most used weapon by time from all weapons. If we combine the facts that the odds of obtaining a knife are low and it is the most used weapon by the player, it can be assumed that in general the demand for knives is high, resulting them being expensive. They use the float value and pattern value. There are multiple knife series as shown in the table 1.4.

Table 1.4: Summary of knives in each series.

Original	Horizon	Classic	Shattered web	Other
Bayonet	Navaja	Classic	Nomad	Bowie
Flip	Stiletto		Paracord	Butterfly
Gut	Talon		Skeleton	Falchion
Karambit	Ursus		Survival	Huntsman
M9 Bayonet				Shadow Daggers

Some cases contain just one type (for instance CS20 case contains only classic knives, or in case of the series other, each knife was introduced in separate case, but then were grouped together in other cases). Every knife can also have finish (skin). There are currently three types of finishes in the game. Original finishes, containing thirteen different skins including vanilla¹¹, Chroma finishes, containing six different skins and Gamma finishes also containing six different skins.

Gloves work similar, except they are only divided into series, because each series have only one type of finish. This could obviously change in the future. There are thirty-eight cases in total currently in the game, but it is expected that more will come.

As already mentioned, gloves work on the similar basis as knives. They can be only obtained from cases and when they are opened the pattern value and float value is generated. At the time of writing this thesis there are currently three glove sets, with each set containing 6 types of gloves with 4 finishes (skins) each. When a player holds a weapon, besides the weapon he can see player models gloves and a little bit of arms, depending on

¹¹ Vanilla knife is a knife without a finish.

players view model¹². This results in gloves being the only item that is visible for the player at all times of playing the game.

1.5 Items outside of Steam market

This entire system works due to the Steam market and trading system. However, the one issue with investing in skins is that when an investor wants to withdraw. Since it is not possible to withdraw money from Steam funds, player cannot just sell the skin on Steam market. As already mentioned, Steam funds can be only used to buy other software on Steam, or to purchase an item on Steam market or directly in the game.

If a player wants to withdraw money, he must sell it to another player. The most general way is to use an external marketplace as a provider. It works on the same basis as Steam market, the buyer deposits funds to the marketplace and then purchases the selected skin from the seller. The seller then receives funds on the website that can be withdrawn. Most websites use peer to peer (or P2P). P2P allows buyer to directly purchase the skin from the seller, without using any third-party account, or bots (Rummler, 2023). To make this method work, both sides (buyer and seller) must provide their API key to the marketplace so that they can track the items via Steam API. Steam API contains public methods that can be accessed from any application capable of making an HTTP request (Steamworks, 2023).

Since the main purpose of CS:GO skins are cosmetics changes and not an investment, there are certainly risks when it comes to investing. There is no guarantee when the developer Valve changes the system (for instance disables trading). Besides that, the whole market is not that large, so there is a potential of whales. Especially for items that are no longer available to obtain directly from the game and the supply cannot rise. In terms of investing a whale is an influential investor that moves with the market by buying or selling activity (fidelity, 2023).

1.6 Investing

Whether can be CS:GO items considered as an alternative investing instrument, it is necessary to understand what investing is and what is considered as a traditional investment instrument.

Investment is a purchase of an asset with the aim of evaluation. There are criteria of decisions when it comes to investing such as returns, risks, liquidity, wealth and more. There are many strategies investor can approach when he makes a portfolio. Portfolio is a collection of assets. The main purpose of making a portfolio is diversification, as it

¹² View model is a point of view of the character that player controls. In CS:GO it is the position of hands holding a weapon.

significantly lowers the risks of the investment. Since there are random variables and variances this problematic could be for a separate thesis (Munk, 2018).

1.6.1 Traditional investment instruments

A traditional investment instrument is a well-known asset that is well recognized in the investing world. Among the traditional instruments are stocks, bonds and other fixed income assets and also cash (Twelveest, 2022).

Investing in stocks is a purchase of shares of the ownership in a company. Investor can either buy shares with the expectation of growth or may receive payments from these shares called dividends. There are over 4000 companies listed on the US stock exchange (TheGlobalEconomy, 2019). When an investor wants to invest into the US market overall, instead of buying individual stocks it is available to purchase an Exchange-traded fund (ETF). ETF copies the price of a selected asset or index. The most recognized index is S&P 500 which consists of five hundred biggest companies in the USA. The market cap is up to 80 % of the total US stock market capitalization (Slickcharts, 2023), allowing this index to measure the performance of the whole US market.

Bonds are debt securities where investor lends money to government or organisations in an exchange for regular interest payments in a set period (Twelveest, 2022).

The last traditional investment instrument is money. Money itself will earn no return or a little return when put in savings account or other financial instruments (Twelveest, 2022).

1.6.2 Alternative investment instruments

Since there is no official definition of alternative investment instrument it is usually accepted that any asset that does not fit in to traditional investment instrument is considered alternative. Since they are not that well known as traditional, they tend to be more risky and less liquid. That is obviously compensated with higher expected return, but it does not have to be always that case. Among the alternative investment instruments are private equity (or venture capital), commodity and collectibles (Twelveest, 2022).

Private equity is like investments to stocks, except the companies are not public, but private so they are not available on public exchange. Venture capital is providing funds to a company in early stage to help it grow. Since both methods are not publicly available not every investor can use those methods.

Commodities are mostly physical goods, divided into two categories, raw commodities such as oil, gold, silver and soft commodities like sugar, wheat, or coffee. Commodities are mostly used as an input of production in other goods or services. Besides that, new commodities are declared, such as bitcoin. US Commodity Futures Trading Commission (CFTC) stated that digital assets like Bitcoin, Ethereum and Tether are commodities (Adejumo, 2022). The price of commodities is determined by supply and

demand like equity. However, they do not provide more potential growth to the company, like stocks and shares.

Collectible can be any item that is considered valuable for the collector. The investor purchases the item with the expectations of future growth in price, due to limitations in supply or growing popularity of the item. The price is also determined by the current supply and demand. Among the examples of collectibles are coins, furniture, books, cars, cards, or CS:GO items.

1.6.3 CS:GO items as an investment instrument

In the previous chapter we found out that CS:GO items fall into category collectibles as an alternative investment instrument. Obviously, some items do not meet those criteria like name tag or StatTrak swap tool. However, as mentioned in previous chapters most items come from cases or collections that are available for a limited time which limits the total supply. Growing popularity for every item is not guaranteed and also could be influenced by the usage of best players, but that could be a topic for another thesis (Mykola, 2022). Due to the growing player base of CS:GO it would make sense that the overall demand for skins and items grows. During the writing of this thesis, on 11.2.2023 CS:GOs player base reached new all-time high of 1 320 219 players at one moment, for comparison the peak of players on July 2021 was 763 523 players at one moment (SteamCharts 2023).

2 Empirical part

The main goal was presented in the beginning, and it is to find out, whether there is a relation between the change of prices of selected CS:GO items and traditional or more known investment instruments. To achieve this goal, ordinary least square (or OLS) method will be used for a creation of two regression models. For the construction of the models and further analysis programming language R in R studio will be used.

Before we retain data, it is necessary to determinate what variables will be used in the models. In the first model the depended variable is a CS:GO item called Operation Breakout Weapon Case. The theoretical part introduced all possible CS:GO items to choose from. The reasons behind the selection of Operation Breakout Weapon Case are the large available quantity during the selected period and high volume of sales. More expensive skins are not sold every day, and some skins are not even sold on the Steam marketplace at all, that leads to price instability of these items. The reasoning behind the choosing of this case, is that it only contains butterfly knives as an exceedingly rare item. To this date, butterfly knives are most popular and most expensive knives, resulting in high demand even for the case (Patterson, 2023). The depended variable in the second model is an item called AK-47 Redline in Field-Tested condition, which is illustrated in Appendix A. This skin is one first in the game and is one of the best-known skins in the game, which is the reasoning of this particular selection (ProSettings, 2023). The main reason is once again the volume of sold units since other forms of selection would not work well. Those two will make a great combination for an analysis to achieve the main goal of this thesis.

Among the independent variables are S&P 500, gold, and bitcoin as they are more known options in investing area in comparison to any CS:GO item.

2.1 Data acquisition

Before we start with the model, we must acquire data for each variable in the selected period which is between January 1 of 2020 to December 31 of 2020.

The S&P 500 is one of the most popular investment instruments, due to its long-term profitability. Investors can buy ETFs that copy the S&P 500 index however, it is still the best solution to choose the index over some specific ETF. Data were acquired from the NASDAQ in US dollars.

In comparison to S&P 500, gold is considered as more stable instrument, a necessary component for this analysis. Data were acquired from COMEX again in US dollars.

On the other hand, Bitcoin is considered as less stable and more volatile commodity. It is still quite new investment instrument. Data for Bitcoin were acquired from coinmarketcap.com, once again in US dollars as one of the most reliable websites for cryptocurrencies.

And lastly our depended variables Operation Breakout Weapon Case and AK-47 Redline. Since there is a graph for each item on Steam market. It is possible to retrieve data from this graph into json file via the price history page on Steam market. Then the data for both items were transformed from json file to csv.

Since NASDAQ and COMEX are not open on weekends and holidays data corrections were necessary. Additional dates when exchanges were closed, were removed from the Bitcoin, AK-47 Redline and Operation Breakout Weapon Case. When inspecting the data, due to the prices of Operation Breakout Weapon Case, three decimals were chosen to make the data ready for further analysis.

To summarize the data acquirement a summary statistic was created in table 2.1. Due to the previously mentioned changes of data all variables must have the same sample size which is 253. Besides that, we can observe that the difference between min values and max values of our depended variables and independent variables are quite significant. However, that should not cause issues in our analysis.

Table 2.1: Summary statistics of the acquired data.

Variable	N	Mean	St. Dev.	Min	Max
Case	253	0.737	0.0457	0.291	1.088
Skin	253	15.069	0.636	13.365	18.100
SaP500	253	3,217.856	101,905.100	2,237.400	3,756.070
Bitcoin	253	11,065.950	17,690,285	5,002.580	28,841.570
gold	253	1773.499	19,421.940	1,469.300	2,045.500

2.2 The first model

2.2.1 Preparation of the first model

Before we begin with the first OLS model with the depended variable Operation Breakout Case, it is necessary to check whether the data contain stochastic trend or not. To check the stochastic trend a stationarity test (or unit root test) will be tested. This is an important check since, with non-stationary data we cannot produce reliable results from further analysis.

There are many tests that analyse the stationarity of a time series. Since we need to have every time series stationary, every variable will be tested separately. The chosen method is Augmented Dickey-Fuller test (ADF) with the results in table 2.2. The results are divided into two columns where each value has a represented its p-value from the separated test.

Table 2.2: Results of the ADF test.

Variable	p-value
S&P 500	0.5527
Operation Breakout Case	0.3984
Bitcoin	0.99
Gold	0.7433

Since the null hypothesis states that the time series is non-stationary and in each test the p-value is well over 0.05, it is obvious that we fail to reject the null hypothesis and that the data are not stationary. To make the data stationary we can difference them. But before we do that, we can perform a cointegration test.

In case the time series are cointegrated it would not be necessary to difference them, but rather convenient to use OLS. Cointegration exists when two or more variables develop in a synchronise way in time. The goal of the cointegration test is to find out whether the relationship is statistically significant, or whether it could be just based on randomness. The cointegration is checked with the Johansen test with the key information presented in table 2.21, where r indicates how many cointegrated time series there are. To check the existence of cointegration we use the results from the test column in comparison to another column with selected confidence. In our case we will use the 5 % confidence column. The test value is only greater when r is equal to 0. This tells us that there is more than zero, but only up to one cointegration in our data set at maximum.

Table 2.21: Results of the Johansen test.

r	test	10 % conf	5 % conf	1 % conf
3	1.81	6.50	8.18	11.65
2	7.91	12.91	14.90	19.19
1	16.18	18.90	21.07	25.75
0	40.01	24.78	27.14	32.14

That is obviously not enough to begin with the OLS and to perform the first regression. This means there is no other option than to difference the data. Before we begin with the regression a visualisation of the newly created data was made. However, since we know that there is a high difference between the min and max of our variables to make the plot clearer an approach of the scaling of the y axis was used. The scaling was done logarithmically and since the variable case has values lower than one, scaled values of this variable were multiplied by minus one to match other variables., with the results shown in the figure 2.1.



Figure 2.1: Plot of newly created data with scaled y axis.

2.2.2 Creation of the first model

After that we have finally prepared our data and we can perform our first regression with the results in table 2.22.

$$\Delta Case = \alpha + \beta_1(\Delta Sap500) + \beta_2(\Delta Bitcoin) + \beta_3(\Delta Gold) + \varepsilon \quad (2.1)$$

Table 2.22: Results of the first regression.

Depended variable:	
	diffCase
diffSap500	-0.00002 (0.00003)
diffBitcoin	0.00000 (0.00000)
diffGold	0.0001 (0.0001)
Constant	0.002 (0.002)

Observations	252
R ²	0.010
Adjusted R ²	-0.002
Residual Std. Error	0.026 (df = 248)
F Statistic	0.856 (df = 3; 248)
Note:	*p<0.1; **p<0.05; *** p<0.01

At first, we can look at the coefficient of determination or R^2 . This value ranges from zero to one. The higher the value is the better the prediction of values of depended variable can be. In our case the value is 0.01 which is in general considered as low, however, it implies that the coefficient of determination is higher than zero. Since with more variables the R^2 tends to grow, an adjusted R^2 comes in to solve that issue. The results show negative adjusted R^2 which indicates that other variables do not add value to the model. After that, another important value is from the F Statistic. If we look at the p-value of the F statistics it is 0.464. The null hypothesis of this test is that there is no statistically significant relationship between the depended variable and other variables. With our p-value of 0.464 we fail to reject the null hypothesis. However, before we conclude the results of the model, it is necessary to performs tests of OLS method to make sure our model does not have any issues.

2.2.3 Testing of the first model

The first test will find whether the residuals of the model are heteroskedastic or homoscedastic. That will be checked with the Breuch-Pagan test. Before we perform the test, a spread level plot is made and is presented in figure 2.2.

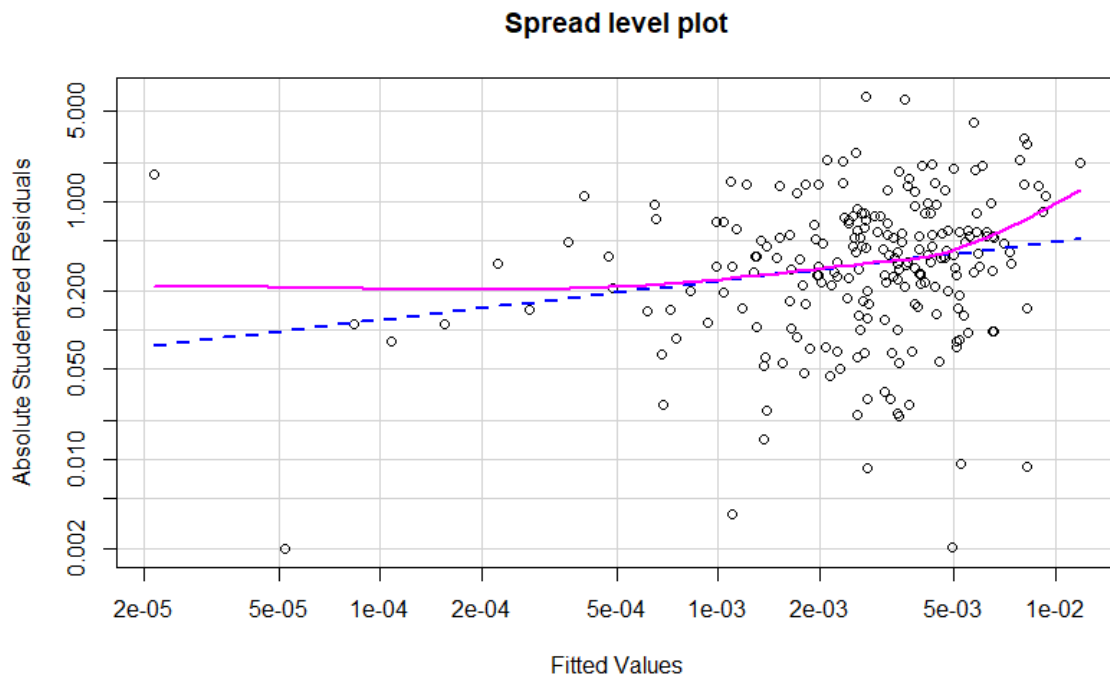


Figure 2.2: Spread level plot.

Since we want to have the conditional variance of a residuals constant or in other words to have the model homoscedastic, the drawn line in figure 2.2 should be ideally horizontal. The line looks promising however, the Breuch-Pagan test will be run anyways with the results in the table 2.23.

Table 2.23: Results of the Breuch-Pagan test.

data:	reg1	
BP = 0.93779	df = 3	p-value = 0.8163

The important indicator from the test is the p-value with the value of 0.8163. Since the null hypothesis in this test is that the model is homoscedastic, we fail to reject the null hypothesis and can assume that the residues homoscedastic.

Moving on to the second test of normality. This test checks whether the residues in model come from normal distribution, it is not a required for the OLS model being unbiased, inconsistent, or inefficient, but rather for further hypotheses. However, even without the normality, the OLS estimator is best linear unbiased estimator (or BLUE). To determine whether the residuals come from normal distribution we will use a quantile-quantile plot (or QQ plot), which is presented in figure 2.3.

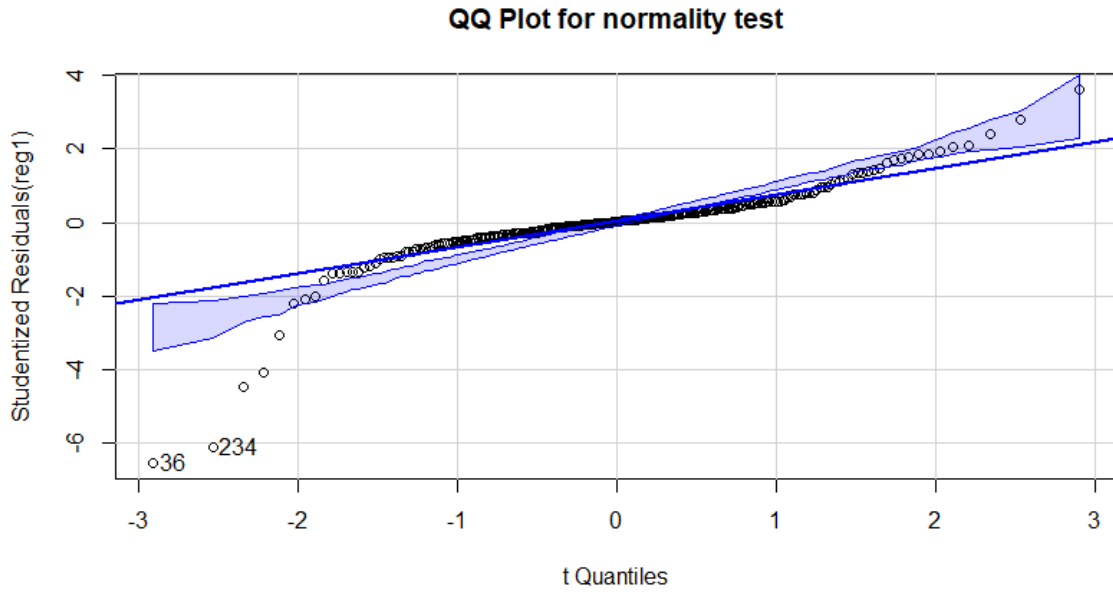


Figure 2.3: QQ plot.

Ideally the points should follow the blue line or be inside the blue area. The figure 2.3 tells us that the most of residues in the model are near our criteria, nevertheless we can also see some extremes like points that are labelled as 36 and 234. Based on that we might assume that the distribution is not normal. Since most tests of normality are sensitive to minor extremes in smaller sample size, no test to check normality will be run.

The last test will check whether there is collinearity or multicollinearity in our model. Collinearity refers to a situation where two explanatory variables are linearly depended. Because then it might be possible to linearly predict one variable from the other variables with almost perfect accuracy. Multicollinearity is then a situation where there are more than two explanatory variables linearly depended in a multiple regression. It also causes that the standard prediction error is higher and that leads to higher p-values in t tests.

The solution to evaluation of the presence of collinearity or multicollinearity is a variance inflation factor (or VIF). This indicator measures the extent to which the variance of estimated coefficients is enlarged in comparison to a situation when the predictors would be mutually independent. VIF is presented in equation 2.2, where j is a selected variable.

$$VIF_j = \frac{1}{1 - R_j^2} \quad (2.2)$$

The interpretation of the results depends on the context. If the result is one, there is no sign of any multicollinearity at all. The higher the indicator is the higher the multicollinearity is. The value of 10 or more is considered as being problematic since the

degree of multicollinearity is serious. However, in some cases even a value of 2 could be a problem. The results of the VIFs indicators are presented in table 2.24.

Table 2.24: Results of VIF indicator.

diffSap500	diffBitcoin	diffGold
1.061774	1.005416	1.056754

As we can see from the results all variables are close to value one, which strongly indicates that there are no issues with collinearity. With these results we can conclude that there is only small collinearity or multicollinearity in our model that does not cause any issues. That allows us to continue to evaluation of our results of the first OLS model.

2.2.4 Evaluation of the first model

The first model with the dependent variable Operation Breakout Case statistically suggests that there is no relation between this loot box and the selected variables. These results can be appealing from an investment point of view since its development is unique in comparison to the selected investment instruments. On one hand it could be risky to invest into an instrument that develops completely different from other investment instruments. On the other hand, it might have a good potential for a riskier portfolio to more diversify investors assets.

2.3 The second model

2.3.1 Preparation of the second model

The second regression model will have the depended variable AK-47 Redline, which is a skin of classified rarity on the AK-47 rifle. Since we already know from the previous chapter that the data do have a stochastic trend or are not stationary, we can skip this test. That leads us straight to the cointegration test. Once again, a Johansen test will be performed, and the results of the test are presented in table 2.31.

Table 2.31: Results of the Johansen test.

r	test	10 % conf	5 % conf	1 % conf
3	2.15	6.50	8.18	11.65
2	8.27	15.66	17.95	23.52
1	22.10	28.71	31.52	37.22
0	47.75	45.23	48.28	55.43

The findings of cointegration are done by the checking the test column in comparison to another column. Again, our selection is the column with confidence of 5 %. The value in test column in the row where $r = 0$, is lower. This means that there is no sign of cointegration at all. This leads to no other option than to difference the data. After that, a plot with logarithmic scaling is made in figure 2.4 to clearly show our variables.

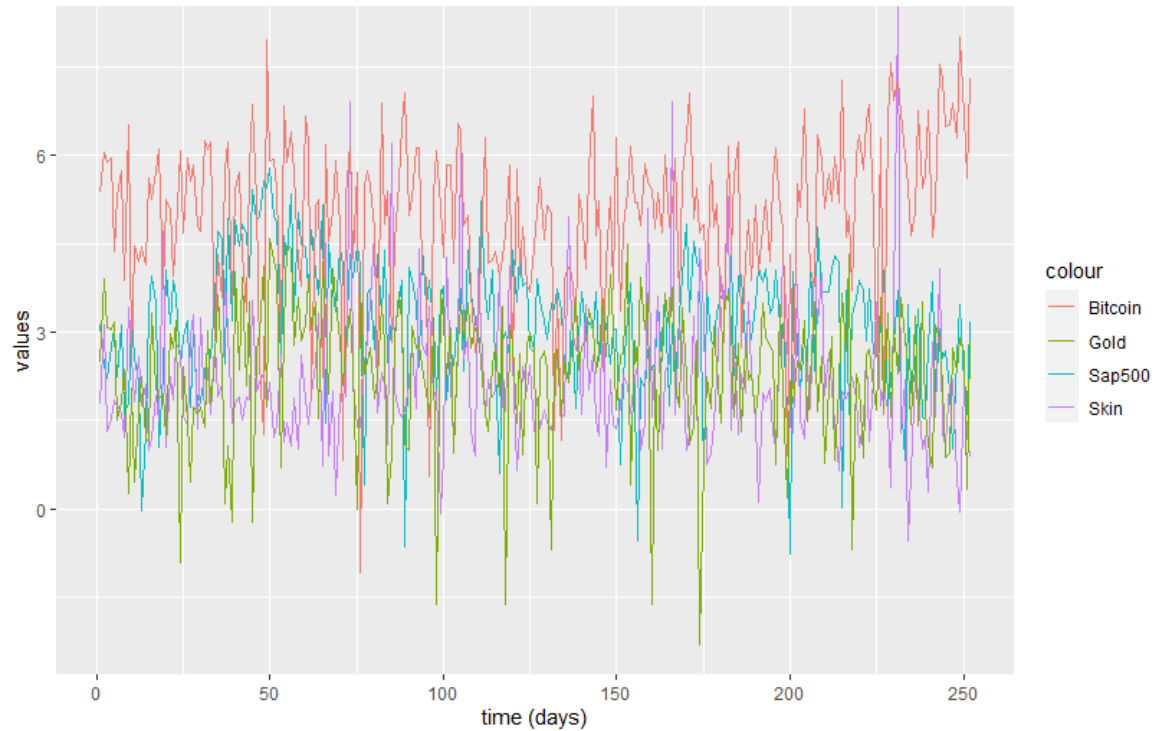


Figure 2.4: Plot of differentiated data in the second model.

2.3.2 Creation of the second model

With that we can move on to the OLS method. The first step is to perform the regression with the depended variable being the AK-47 redline skin. The results are presented in table 2.32.

$$\Delta Skin = \alpha + \beta_1(\Delta Sap500) + \beta_2(\Delta Bitcoin) + \beta_3(\Delta Gold) + \varepsilon \quad (2.3)$$

Table 2.32: Results of the regression.

Depended variable:	
	diffSkin
diffSap500	-0.0002

	(0.0003)
diffBitcoin	0.0001** (0.00003)
diffGold	0.001 (0.001)
Constant	0.008 (0.017)
Observations	252
R2	0.035
Adjusted R2	0.023
Residual Std. Error	0.0259 (df = 248)
F Statistic	2.995** (df = 3; 248)
Note:	*p<0.1; **p<0.05; *** p<0.01

We can see from the results that this time the coefficient of determination is higher, however, it is still far from one. This time the adjusted R^2 is positive and has a value of 0.023. That indicates that the model has a better fit, nevertheless, it is still low. Continuing to the F statistic, we can see that the p-value is below 0.05. As mentioned previously the null hypothesis of this test is that there is no relationship between the depended variable and other variables. With these results we can reject the null hypothesis and accept an existence of at least one relationship between the depended variable and independent variables. Subsequently if we look at the p-values of each variables t-tests we can see the same result for a variable Bitcoin, where the significance is on p-value of 0.05. This could confirm an existing relation between the development of price of bitcoin and the AK-47 Redline skin in Field Tested condition. But before we can conclude this it is necessary to perform tests of the OLS method.

2.3.3 Testing of the second model

The first test will find whether the model is heteroskedastic or homoscedastic. At first the spread level plot will be made and presented in figure 2.5 for a visual perspective.

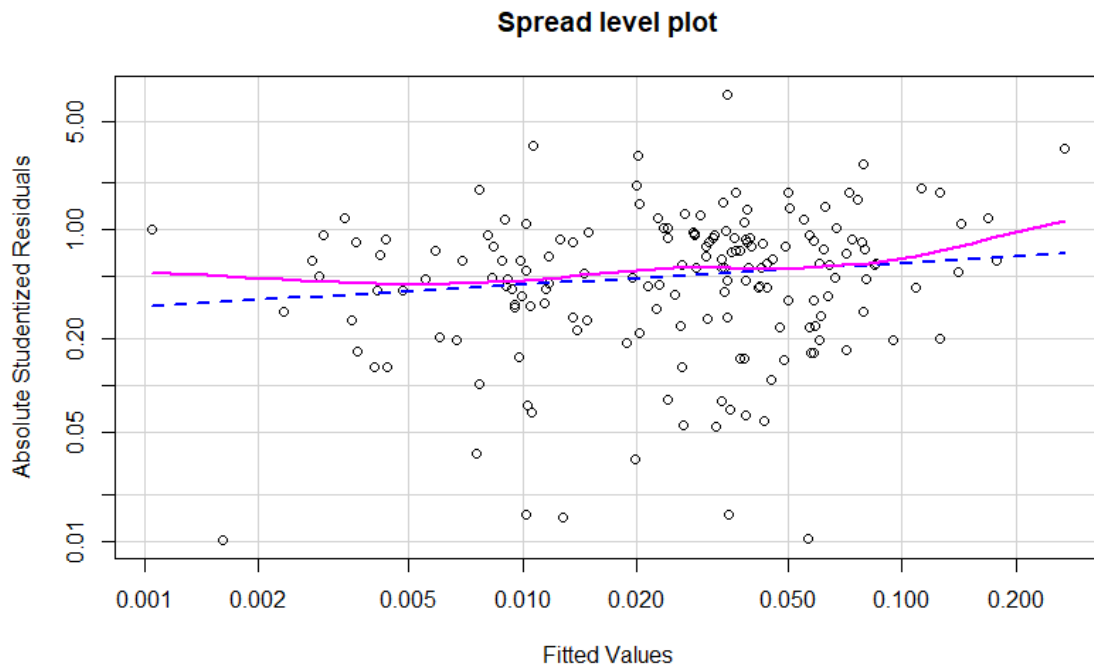


Figure 2.5: Spread level plot.

We can assume from the figure that the model is homoscedastic since the line is almost horizontal. To make sure homoscedasticity is in place, we will use the Breuch-Pagan test with the results in table 2.33.

Table 2.33: Results of the Breuch-Pagan test.

data:		
reg2		
BP = 2.7906	df = 3	p-value = 0.425

This time the p-value as the important indicator values at 0.425. That indicates that the second model is not heteroskedastic. With these results we can move onto the second test.

The second test will check whether the residues in the model come from a normal distribution. As mentioned previously, this is not a requirement for the OLS model to be BLUE so no test will be run. However, we will create a QQ plot to visualise the situation with the results in figure 2.6.

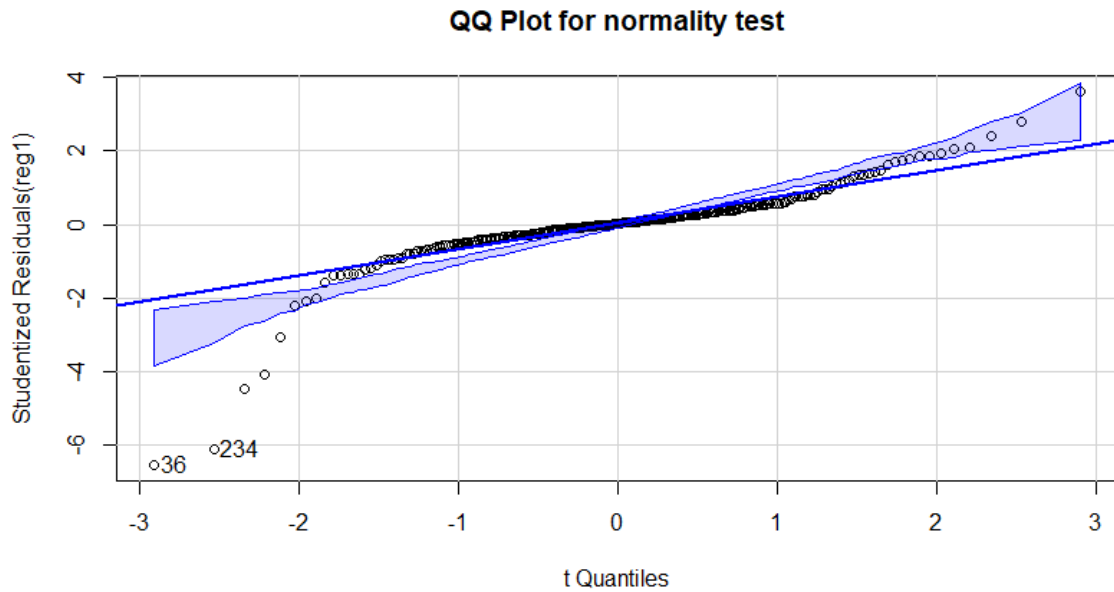


Figure 2.6: QQ plot.

Once again, we look whether the points follow the blue line or are inside the blue area. We can see comparable results as in the previous model in figure 2.3, where we assumed that most of the points seems to follow our criteria, however, due to the extremes at points 36 and 234, we estimated that the distribution is not normal. With that we can move onto the last test of the second model.

The last test before the evaluation of the second model is the test of collinearity and multicollinearity. The reason for this testing has been explained in the previous testing of collinearity and multicollinearity in the chapter 2.2.3 in the testing of the first model. Once again, a VIF indicator will be used to evaluate this potential problematic. The results of the VIF indicator are shown in table 2.34.

Table 2.34: Results of VIF indicator.

diffSap500	diffBitcoin	diffGold
1.061774	1.005416	1.056754

We can see from the results that values of variables are all close to one. That strongly indicates that the collinearity or multicollinearity is not an issue in this model. With these results, we can move onto the evaluation of the second model.

2.3.4 Evaluation of the second model

The depended variable in the second model is AK-47 Redline in Field-Tested condition. The variable diffBitcoin or differentiated Bitcoin shows a result, which suggests a relation between development in price of Bitcoin and the selected. Nevertheless, it does not mean that this skin does not have an investment potential. The relation could be existing

only for the analysed period and from an investment perspective it would also be important to look at things like coefficient of the differences, stability and more. This result also shows an important warning that it is necessary to look at individual CS:GO items and not to skins or items as a whole.

Conclusion

The introduction addresses the current growth of popularity for e-sports and for the collectable items in video games. It highlights the video game CS:GO where it is possible to trade the in-game items between players and buy or sell them on Steam market that is based on supply and demand. This allows to further investigate the problematic and to select the goal of this thesis which is to find out whether there is a relation between these items and more traditional investment instruments, since the results might help to understand whether the CS:GO items do have an investment potential or not.

Theoretical part introduces the game itself and the items that can be obtained in the game. Each item is examined in detail for a better understanding of the potential of every item. Furthermore, issues that are related to this topic are pointed out such as the limitation of the Steam market that is capped to \$2000. This could be a serious restriction for investors that would like to invest in larger quantities however, it could be bypassed by the ownership of multiple accounts, but it certainly makes the investments less convenient and less secure. Last part of the theoretical part briefly introduces investment instruments for a better understanding in the empirical part.

The analysis of the goal of this thesis is done in the empirical part, where two regression models are made. Selected period that is examined in both models is the year 2020. The first model analysis the Operation Breakout Case which is a loot box from CS:GO that is high-volatile and contains the most popular knife in the game. Results of this model do not show a statistical sign of a relation between the loot box and selected investment instruments which were: S&P500, Gold and Bitcoin. That might open an investment opportunity for a more diversity in portfolio, however, bear in mind that the selected period is only one year, and the analysis does not include the development of the case itself which might not be profitable.

The second model analysis a skin on AK-47 called Redline in Field Tested condition, which is one of the most known skins in the game. Second model provides different results since there is a statistically significant relation between this skin and Bitcoin. Nonetheless, it does not signify that the skin does not have an investment potential, because other things play role and were not analysed such as the coefficient of the differences, impacts of external phenomena on the development and more. However, the results of the second model showed an important finding, that it is important to analyse the items individually, hence we cannot look at the CS:GO items, or just cases or skins as a whole.

In conclusion, results of this thesis indicate that price of a selected item from CS:GO can have unique development (the case of loot box Operation Breakout Case), with expectation of this existence in more items in the game. However, it also points out that at least one item (AK-47 Redline, Field Tested) does have an existing relation in development of price to Bitcoin, once again with expectation that more items might also have a similar

development to existing investment instruments. Consequently, more research for other items in CS:GO would take a better picture of this topic.

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Appendices

Appendix A: AK-47 default weapon vs AK-47 Redline Skin



On the left side is the default weapon AK-47 without skin. On the right side is the Redline skin on the same weapon in Field-Tested condition without any stickers.